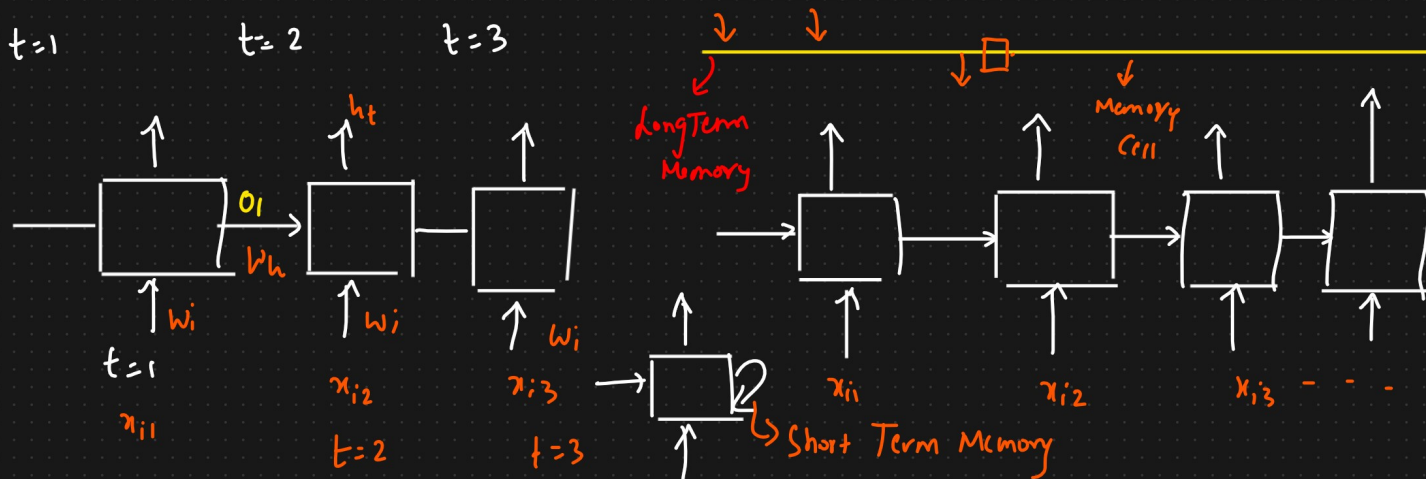
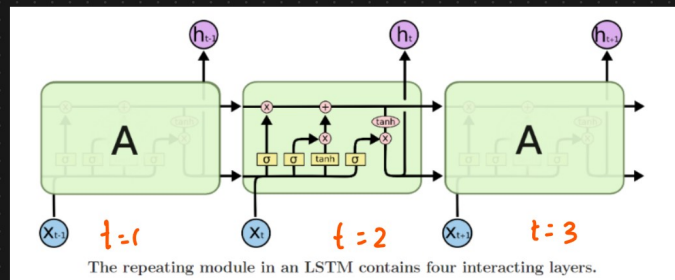
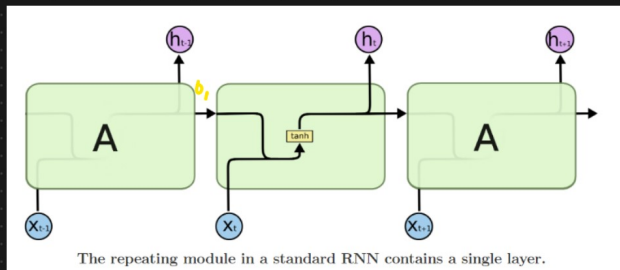


Basic Representation of RNN And LSTM RNN

LSTM RNN

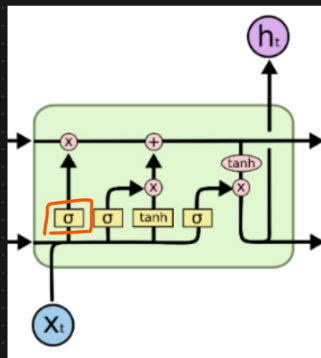


LSTM RNN
 ↗ Long Term Memory
 ↘ Short Term Memory

Conveyance But : Luggages

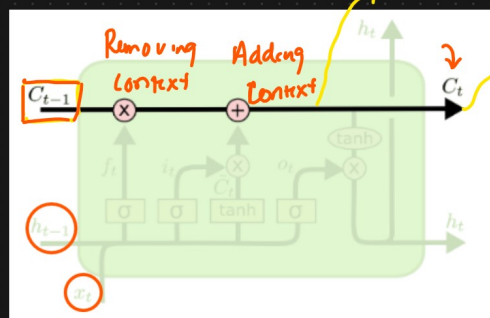
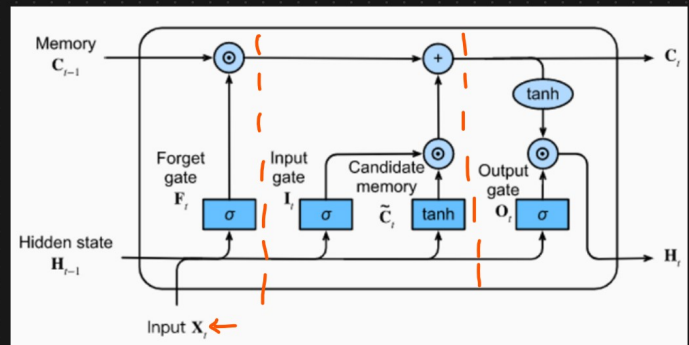


LSTM Architecture



Basic Architecture

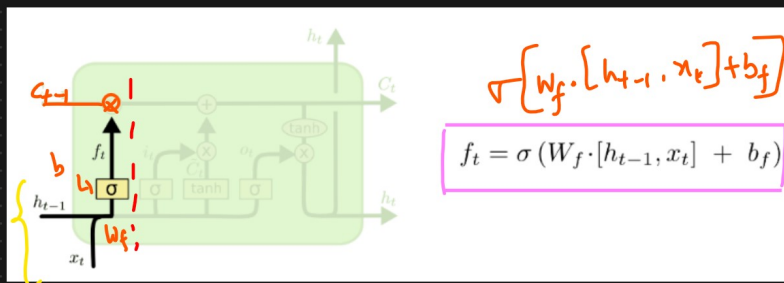
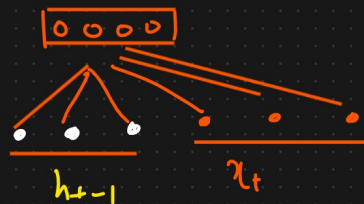
LSTM RUN



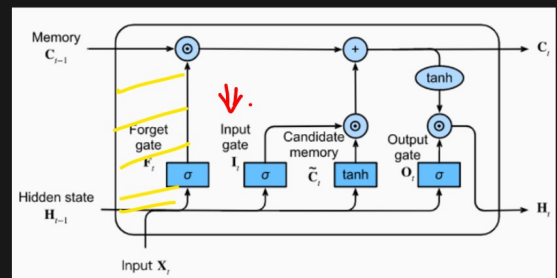
Combining 2 vectors

$$h_{t-1} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$x_t = \begin{bmatrix} 2 & 3 & 4 \end{bmatrix}$$



$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$



Forget gate

h_{t-1} = Hidden state of previous time stamp

x_t = Word passed as i/p in the current time stamp

Text: $x_{11}, x_{12}, x_{13}, x_{14}$
Next Word: y_{15}

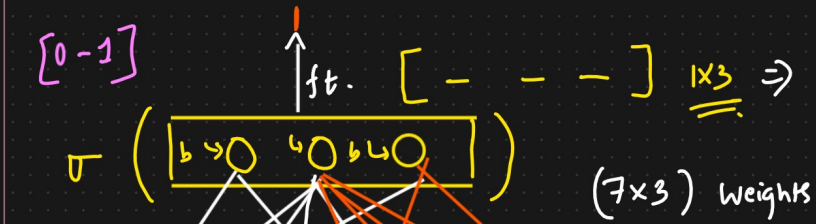
$$\begin{bmatrix} 0 & 2 & 4 & 1 \end{bmatrix} \quad \begin{bmatrix} 4 & 5 & 12 \end{bmatrix} \quad \dots$$

x_t x_{t+1}

$$h_{t-1} \begin{bmatrix} 1 & 2 & 4 \end{bmatrix} \quad c_{t-1} \begin{bmatrix} 4 & 2 & 1 \end{bmatrix} \Leftarrow$$

$3d$ $3d$

$$c_{t-1} \otimes c_t =$$



$(1 \times 7) \times (7 \times 3)$

(1×3)

$f_t = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \leftarrow \text{Complete Sentence}$

Context changed

① $c_{t-1} = \begin{bmatrix} 6 & 8 & 9 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \leftarrow \text{Removing all the previous Context}$

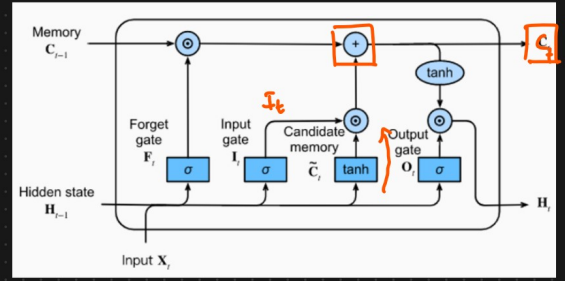
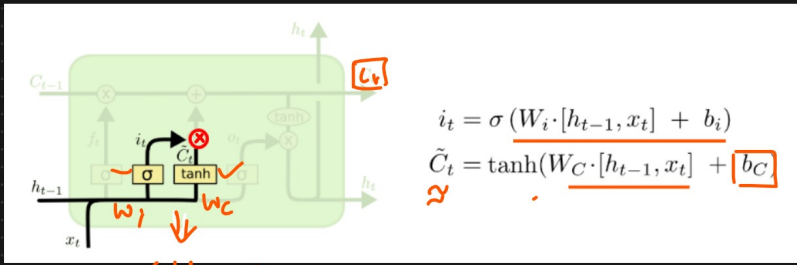
$f_t = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$

② $c_{t-1} = \begin{bmatrix} 6 & 8 & 9 \end{bmatrix} \otimes \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 8 & 9 \end{bmatrix}$

③ $c_{t-1} = \begin{bmatrix} 6 & 8 & 9 \end{bmatrix} \otimes \begin{bmatrix} 0.5 & 1 & 0.5 \end{bmatrix} = \begin{bmatrix} 3 & 8 & 4.5 \end{bmatrix}$

Conclusion : Based on the context $\rightarrow$ Forget gate will let go some information or will not let go some Info {Forgetting}.

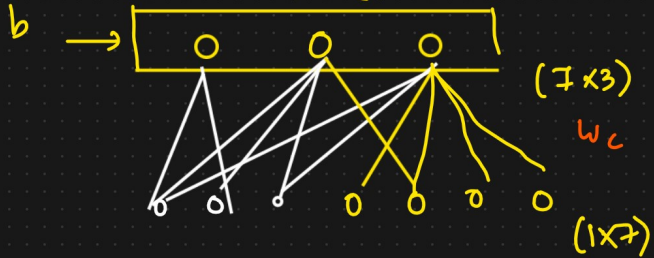
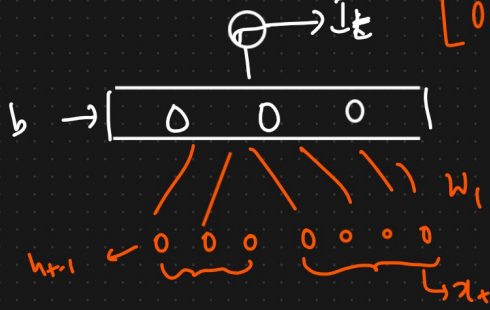
② Input Gate And Candidate Memory



Adding Info

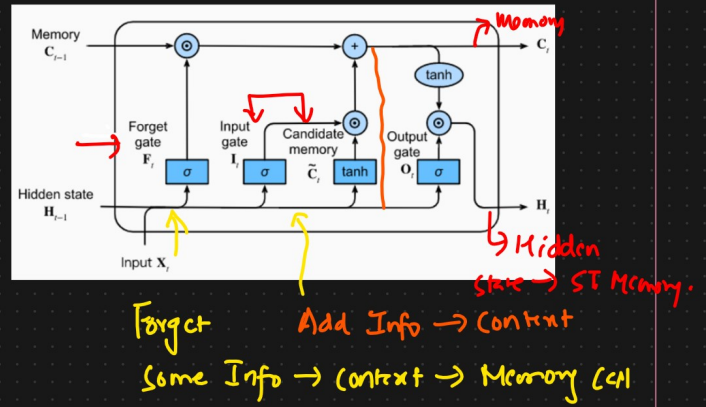
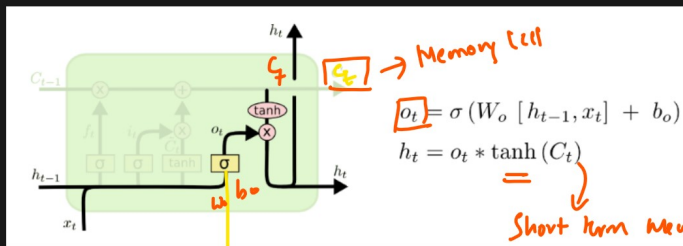
$$I_t = [2 \ 4 \ 6] \text{ --- } (4) \text{ --- } \underset{\uparrow}{C_t} [0 \ 2 \ 0] \Rightarrow \underline{\underline{\text{I/p Gate}}}$$

$$\begin{bmatrix} 0 & 8 & 0 \end{bmatrix}$$



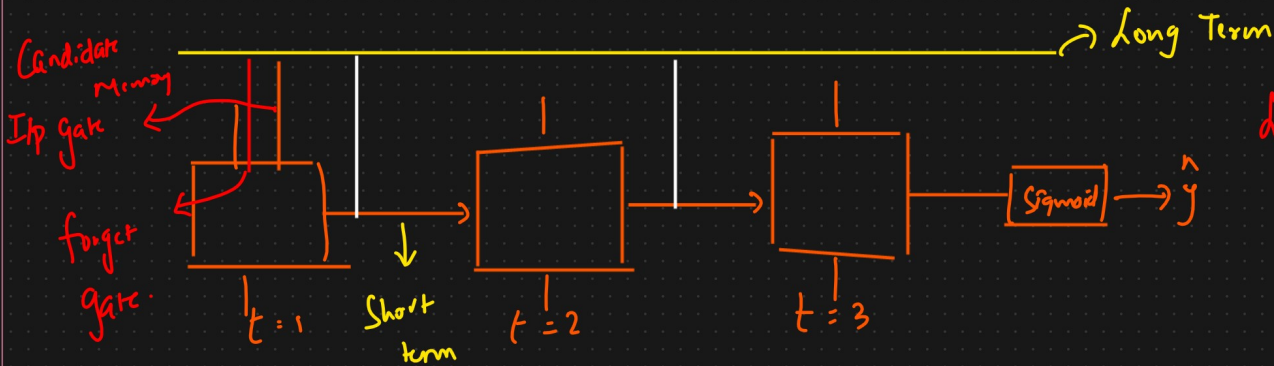
Context = If any information needed to be added it the memory
 $C_{t-1} \rightarrow$ The information will be added

Output gate LSTM RNN



h_{t-1} → Long Term Memory

o_t → h_t ⇒ hidden state → Short term Memory



$[w_i, w_c, w_o] \rightarrow$ Updating ← Back Propagation

GRU RNN ⇒ LSTM Variant